

LC 529: Minesweeper

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Legend

- 'M' → unrevealed mine
- 'E' → unrevealed empty square
- 'B' → reveals a blank square with no mines in all 8 directions
- 'X' → revealed mine
- **click array** → represent next click, among M and E

Rules

- **i)** If 'M' clicked then game over, change it to 'X'
- **ii)** If 'E' clicked then change to 'B' and all adjacent unrevealed square are revealed successfully
- **iii)** If 'E' click with atleast one adjacent mine, then change to '1' to '8' depending on num of adjacent mine
- **iv)** return final board

Cases

Case 1	M → X (game over)
Case 2	E → count adjacent mine let us say 1 adjacent mine → '1'
Case 3	E → no adjacent mine E → B, reveal all 8 neighbours → DFS Eg (Click E): E E E → B 1 E E E M → B 1 M E E E → B 1 E Note: DFS stopped as cells become numbers

⚠ DFS expands only when adjacent mine count == 0

C++ Solution

```
class Solution {
public:
    vector<vector<char>> updateBoard(vector<vector<char>>& board,
vector<int>& click) {
        int n = board.size();
        int m = board[0].size();

        int r = click[0];
        int c = click[1];

        if(board[r][c] == 'M') {
            board[r][c] = 'X';
            return board;
        }

        vector<pair<int,int>> dir = {{-1,-1},{-1,0},{-1,1},{0,-1},{0,1},
{1,-1},{1,0},{1,1}};
        queue<pair<int,int>> q;
        vector<vector<int>> seen(n, vector<int>(m, 0));

        // seen ki zarurat nhi , as E se B ya digit change krenge to
already marked seen
        seen[r][c] = 1;
        q.push({r, c});

        while(!q.empty()) {
            int r = q.front().first;
            int c = q.front().second;
            int cnt = 0; // dekhna padega kitne mines he aaju baaju
            q.pop();

            for(auto &it : dir) {
                int nr = r + it.first;
                int nc = c + it.second;

                if(nr >= 0 && nc >= 0 && nr < n && nc < m) {
                    if(board[nr][nc] == 'M') cnt++;
                }
            }
        }
    }
};
```

```

        if(cnt > 0) {
            board[r][c] = '0' + cnt;
        }
        else if(cnt == 0) {
            // to phir bfs krna padega

            board[r][c] = 'B';
            for(auto &it : dir) {
                int nr = r + it.first;
                int nc = c + it.second;

                if(nr >= 0 && nc >= 0 && nr < n && nc < m && !
seen[nr][nc] && board[nr][nc] == 'E') {
                    seen[nr][nc] = 1;
                    q.push({nr, nc});
                }
            }
        }
    }

    return board;
}
};

```